

Operator-Norm LLN for Bounded Random Pre-Channel Products

Run `fa_banach_001`, agent `agent_lane_06`

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Status

Claimed full solution, likely valid subject to human review. The packet proves the conjecture stated after Theorem 1 of arXiv:2504.10033. In fact it proves a stronger statement: the random product converges almost surely in operator norm on $\mathcal{L}(\mathcal{T}_p)$, hence in the strong operator topology of $\mathcal{T}_p$. The proof covers the endpoint $p = 1$.

Source Problem

The source paper is S. V. Dzhenzher and V. Zh. Sakbaev, *The law of large numbers for discrete generalized quantum channels*, arXiv:2504.10033. Theorem 1 of the source proves, for $1 \leq p \leq 2$, that if $A, A_1, A_2, \dots$ are bounded i.i.d. random pre-channel generators on $\mathcal{T}_p$, then

$$e^{A_1 t/n} \dots e^{A_n t/n} \longrightarrow e^{(\mathbb{E}A)t}$$

almost surely in the strong operator topology of $\mathcal{T}_2$, uniformly for t in compact intervals. Immediately after the theorem, the paper states:

It is interesting to obtain the LLN in SOT of $\mathcal{T}_p$ not $\mathcal{T}_2$. This leads us to the following conjecture.

Under the assumptions of Theorem 1,

$$e^{A_1 t/n} \dots e^{A_n t/n}$$

converges almost surely in SOT of $\mathcal{T}_p$ to $e^{\mathbb{E}At}$ uniformly for t in any segment.

semigroup e^{At} such that $\|A\|$ is bounded. Let $A, A_1, A_2, \dots$ be a sequence of random i.i.d. generators. Then $e^{A_1 t/n} \dots e^{A_n t/n}$ converges almost surely in SOT of $\mathcal{T}_2$ to $e^{\mathbb{E}At}$ uniformly for t in any segment.

It is interesting to obtain the LLN in SOT of $\mathcal{T}_p$ not $\mathcal{T}_2$. This leads us to the following conjecture.

Conjecture 2.3. *Under the assumptions of Theorem 2.2, $e^{A_1 t/n} \dots e^{A_n t/n}$ converges almost surely in SC of $\mathcal{T}_p$ to $e^{\mathbb{E}At}$ uniformly for t in any segment.*

3. AUXILIARY LEMMAS

For a Banach space X we denote by X^* the dual space of linear functionals $X \rightarrow \mathbb{K}$. For $x \in X$ and $f \in X^*$ we identify $\langle f, x \rangle := f(x)$.

For any $1 \leq p \leq \infty$ we define p^* by the identity $\frac{1}{p} + \frac{1}{p^*} = 1$. So, for $p < \infty$ we have $\mathcal{T}_p^* = \mathcal{T}_{p^*}$.

For $U \in \mathcal{L}(\mathcal{T}_p, \mathcal{T}_q)$ define the adjoint pre-channel $U^* \in \mathcal{L}(\mathcal{T}_q^*, \mathcal{T}_p^*)$ by

$$\langle U^* y, x \rangle = \langle y, Ux \rangle$$

for any $x \in \mathcal{T}_p$ and $y \in \mathcal{T}_q^*$.

Proof Intuition

The conjecture looks like it should require the geometry of $\mathcal{T}_p$, especially at the endpoint $p = 1$. It does not. The scaling t/n makes every factor a first-order perturbation of the identity, and a coarse block argument removes the noncommutativity.

Split the n factors into m consecutive blocks. Inside one block, whose total time length is about t/m , the product satisfies

$$\prod_{\text{block}} e^{tA_j/n} = I + \frac{t}{n} \sum_{\text{block}} A_j + O(m^{-2})$$

uniformly in n . For each fixed block partition, the ordinary Banach-valued strong law gives

$$\frac{1}{n} \sum_{\text{block}} A_j \longrightarrow \frac{1}{m} \mathbb{E}A$$

almost surely in the operator norm of $\mathcal{L}(\mathcal{T}_p)$. Thus every block is asymptotic to

$$I + \frac{t}{m} \mathbb{E}A + O(m^{-2}),$$

or equivalently to $e^{t\mathbb{E}A/m}$ up to $O(m^{-2})$. Multiplying the m blocks gives an $O(m^{-1})$ total error, and then $m \rightarrow \infty$. This proves operator-norm convergence and therefore the desired strong operator convergence.

Banach-Algebra Theorem

Theorem 1 (Random exponential product LLN in a Banach algebra). *Let $\mathcal{B}$ be a unital Banach algebra. Let*

$$A, A_1, A_2, \dots$$

be i.i.d. $\mathcal{B}$ -valued random variables with

$$M := \text{ess sup } \|A\| < \infty, \quad \bar{A} := \mathbb{E}A.$$

Assume A is Bochner measurable. Then for every $T > 0$,

$$\sup_{0 \leq t \leq T} \left\| e^{tA_1/n} \dots e^{tA_n/n} - e^{t\bar{A}} \right\| \longrightarrow 0$$

almost surely.

Proof. The random variable A is bounded and Bochner measurable, so it takes values in a separable closed subspace of $\mathcal{B}$. The Banach-valued strong law of large numbers gives

$$\frac{1}{N} \sum_{j=1}^N A_j \longrightarrow \bar{A} \quad \text{almost surely in } \mathcal{B}. \tag{1}$$

Work on an event of probability one where (1) holds.

We first record a deterministic estimate. If $b_1, \dots, b_L \in \mathcal{B}$, $\|b_j\| \leq M$, and $0 \leq hL \leq T$, then

$$\left\| e^{hb_1} \dots e^{hb_L} - I - h \sum_{j=1}^L b_j \right\| \leq C_{M,T}(Lh)^2. \tag{2}$$

Indeed, write $e^{hb_j} = I + u_j$. Then $\|u_j\| \leq C_M h$, and the product expansion gives

$$\left\| \prod_{j=1}^L (I + u_j) - I - \sum_{j=1}^L u_j \right\| \leq \exp \left(\sum_{j=1}^L \|u_j\| \right) - 1 - \sum_{j=1}^L \|u_j\| \leq C_{M,T} (Lh)^2.$$

Also $\sum_j \|u_j - hb_j\| \leq C_M Lh^2 \leq C_{M,T} (Lh)^2$, after increasing $C_{M,T}$. This proves (2).

Fix $m \in \mathbb{N}$. Put $N_r(n) = \lfloor rn/m \rfloor$, $r = 0, \dots, m$, and let

$$B_{n,r}(t) := e^{tA_{N_{r-1}(n)+1}/n} \dots e^{tA_{N_r(n)}/n}$$

be the r -th consecutive block, with the convention that an empty block is I . From (1),

$$\frac{1}{n} \sum_{j=N_{r-1}(n)+1}^{N_r(n)} A_j = \frac{N_r(n)}{n} \frac{1}{N_r(n)} \sum_{j=1}^{N_r(n)} A_j - \frac{N_{r-1}(n)}{n} \frac{1}{N_{r-1}(n)} \sum_{j=1}^{N_{r-1}(n)} A_j \rightarrow \frac{1}{m} \bar{A}$$

for each fixed $r = 1, \dots, m$, with the obvious interpretation for $N_{r-1}(n) = 0$. Using (2), uniformly for $0 \leq t \leq T$,

$$\begin{aligned} \limsup_{n \rightarrow \infty} \sup_{0 \leq t \leq T} \|B_{n,r}(t) - e^{t\bar{A}/m}\| &\leq \limsup_{n \rightarrow \infty} \sup_{0 \leq t \leq T} \left\| B_{n,r}(t) - I - \frac{t}{n} \sum_{j=N_{r-1}(n)+1}^{N_r(n)} A_j \right\| \\ &\quad + \limsup_{n \rightarrow \infty} \sup_{0 \leq t \leq T} \left\| \frac{t}{n} \sum_{j=N_{r-1}(n)+1}^{N_r(n)} A_j - \frac{t}{m} \bar{A} \right\| \\ &\quad + \sup_{0 \leq t \leq T} \left\| I + \frac{t}{m} \bar{A} - e^{t\bar{A}/m} \right\| \\ &\leq \frac{C_{M,T}}{m^2}. \end{aligned}$$

The middle term tends to zero uniformly in $t \in [0, T]$, because it is t times a norm-convergent operator.

The full product is the product of these m blocks. The block norms and the norms of $e^{t\bar{A}/m}$ are bounded by constants depending only on M and T . A telescoping estimate therefore gives

$$\limsup_{n \rightarrow \infty} \sup_{0 \leq t \leq T} \|e^{tA_1/n} \dots e^{tA_n/n} - (e^{t\bar{A}/m})^m\| \leq \frac{C_{M,T}}{m}.$$

Since $(e^{t\bar{A}/m})^m = e^{t\bar{A}}$, and m was arbitrary, the desired convergence follows by letting $m \rightarrow \infty$. $\square$

Consequence For The Source Conjecture

Corollary 1. *Let $1 \leq p \leq 2$. In the setting of arXiv:2504.10033, if*

$$A, A_1, A_2, \dots : \Omega \rightarrow \mathcal{L}(\mathcal{T}_p)$$

are bounded i.i.d. random pre-channel generators, then for every $T > 0$,

$$\sup_{0 \leq t \leq T} \|e^{A_1 t/n} \dots e^{A_n t/n} - e^{(\mathbb{E}A)t}\|_{\mathcal{L}(\mathcal{T}_p)} \rightarrow 0$$

almost surely. Consequently, for every $x \in \mathcal{T}_p$,

$$\sup_{0 \leq t \leq T} \left\| \left(e^{A_1 t/n} \dots e^{A_n t/n} - e^{(\mathbb{E}A)t} \right) x \right\|_p \rightarrow 0$$

almost surely.

Proof. Apply Theorem 1 to the unital Banach algebra $\mathcal{B} = \mathcal{L}(\mathcal{T}_p)$. The source assumes the random pre-channel generator is bounded, and its discrete probability-space hypothesis ensures measurability; the expectation used in the source is the Bochner expectation in $\mathcal{L}(\mathcal{T}_p)$. Operator-norm convergence immediately implies convergence in the strong operator topology of $\mathcal{T}_p$. $\square$

Remark 1. *The Banach-algebra theorem is not special to Schatten classes. It applies to bounded i.i.d. generators in $\mathcal{L}(X)$ for any Banach space X , provided the operator-valued random variable is Bochner measurable. The source conjecture only asks for $X = \mathcal{T}_p$, $1 \leq p \leq 2$.*

Verification Notes

The proof uses two standard ingredients: the Banach-valued strong law of large numbers for bounded Bochner-measurable random variables, and the elementary first-order block estimate (2). Human review should focus on the floor-index block passage and on confirming that the source paper’s “bounded random pre-channel” assumption is exactly a bounded Bochner-measurable random variable in $\mathcal{L}(\mathcal{T}_p)$. The source’s discrete probability-space convention appears to give this measurability directly.

The included script `scalar_random_product_smoke.py` is only a sanity check in a finite-dimensional noncommutative toy model; it is not used in the proof.

Novelty Check

Cheap run indexes were searched for 2504.10033, the exact title, SOT of $\mathcal{T}_p$, and discrete generalized quantum channels; no duplicate packet was found. A web search on 2026-06-28 for the exact title and conjecture phrases found the arXiv page and no separate later answer. This packet is therefore recorded as a new claimed full solution, pending human verification.

References

- S. V. Dzhenezher and V. Zh. Sakbaev, *The law of large numbers for discrete generalized quantum channels*, arXiv:2504.10033.
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